Full Stack Assignment – 49

Assignment ID: FSD-49
Title: Rule-Based Discount Eligibility Management System
Duration: 3 Hours

1. Objective

Design and develop a **Rule-Based Discount Eligibility Management System** that determines whether users are eligible for discounts based on configurable rules.
The system must enforce eligibility logic strictly and ensure **data persistence using a database**.

2. User Roles

CUSTOMER

- Register and log in to the system
- View available discount offers
- Check discount eligibility
- Apply eligible discounts

ADMIN

- Register and log in to the system
- Create and manage discount rules
- View discount usage reports

3. Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use JWT-based authentication
- Protect all backend APIs using JWT
- Enforce role-based access control on APIs and frontend views

Discount Rule Management

- ADMIN can create discount rules based on conditions such as:
  - Minimum order value
  - User category or role
  - Usage count limits
- Discount rules must be stored in the database

Discount Evaluation

- System evaluates discount eligibility dynamically at runtime
- Eligibility results must be stored in the database

- Ineligible discount attempts must be rejected

Discount Application

- CUSTOMER can apply only eligible discounts
- Applied discounts must be persisted and reflected in final calculations

4. Business Rule (Critical)

A discount must be applied **only if all configured eligibility rules are satisfied**.
Partial rule matches must not allow discount application.

5. Technical Constraints (Mandatory)

- All discount, rule, and usage data must persist in a database
- Hardcoded or in-memory discount logic is not allowed
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6. Frontend Requirements

- Login and registration pages
- Role-based dashboards for CUSTOMER and ADMIN
- Discount listing and eligibility check view for CUSTOMER
- Discount rule management and usage reporting view for ADMIN

UI design is not evaluated; correctness of rule evaluation is the priority.

7. Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview

- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8. Evaluation Note

Evaluation focuses on **rule correctness, eligibility enforcement, database-backed logic, role separation, and end-to-end execution**.